

DAY 5

Functional Analysis & Mini-Project

Prokka • eggNOG-mapper • HUMAnN3 • KEGG • COG • AMR genes • Mini-project

Why Functional Analysis?

"Knowing who is in a community tells us potential — knowing what genes are there tells us capability — but only function tells us what is actually happening."



Beyond taxonomy

Two communities can have identical taxonomic profiles but very different metabolic potential — and vice versa. Taxonomy alone is insufficient.



Metabolic pathways

Identify complete or incomplete metabolic pathways: carbon fixation, nitrogen cycling, sulfur metabolism, fermentation, antibiotic production.



AMR & virulence

Detect antibiotic resistance genes (ARGs) and virulence factors in communities. Critical for public health and clinical metagenomics.



Ecological function

Understand community roles: primary producers, decomposers, N-fixers, methanogens. Connect microbial function to ecosystem services.

Two Approaches to Functional Annotation

The approach depends on whether you have reads (community-level) or assembled MAGs (genome-level):

Community-Level (Reads)

Use when: You have shotgun reads and want pathway/function abundances across many samples.

Tool: HUMAnN3

Approach:

1. MetaPhlAn4 profile (which species present)
2. Map reads to pangenome databases (UniRef90, ChocoPhlAn)
3. Unmapped reads searched against UniRef90 protein database
4. Output: gene families + pathway abundances per sample

Output format: stratified (per-species contribution) or unstratified (total community).

Genome-Level (MAGs)

Use when: You have assembled MAGs and want gene annotation for each genome.

Tools: Prokka → eggNOG-mapper

Approach:

1. Prokka: predict genes (CDS, rRNA, tRNA) from MAG
2. eggNOG-mapper: map predicted proteins to eggNOG database → COG categories, KEGG orthology, GO terms, PFAM domains
3. Compare functional profiles across MAGs or environments

Output: per-gene and per-MAG annotation tables. Best for detailed genome characterisation.

SECTION 1

Gene Prediction

Prodigal • Prokka • tRNA • rRNA

Prokka: Rapid Prokaryote Genome Annotation

Gene prediction

Prodigal

Predicts all protein-coding sequences (CDS) using dynamic programming. Trained on the input sequence — works without prior knowledge.

Signal peptides

SignalP

Identifies signal peptides for secreted proteins. Optional — skip with `--nosigclip` if speed needed.

rRNA prediction

RNAmmer / Barrnap

Annotates 5S, 16S, 23S rRNA genes. Essential QC for MAGs — high-quality MAGs should have rRNA genes.

tRNA prediction

Aragorn

Identifies all tRNA genes. Complete genomes typically have 36–89 tRNA genes.

Protein function

BLAST / HMMER

Searches predicted proteins against curated databases (UniProt, TIGRFAM, PFAM) for functional annotation.

Output

GFF3, GBK, FAA, FNA

Standard annotation formats: GFF3 (coordinates), GenBank (full annotation), FASTA (protein/nucleotide sequences).

```
prokka --outdir prokka_MAG1 --prefix MAG1 --cpus 8 --kingdom Bacteria MAG1.fa
```

SECTION 2

Functional Databases & Tools

eggNOG • KEGG • COG • MetaCyc • CARD

Functional Annotation & Databases

eggNOG-mapper

Maps proteins to eggNOG orthologous groups. Fast (DIAMOND) or sensitive (HMMER) modes.

Output columns per gene:

seed_ortholog	Best hit in eggNOG database
COG_category	Cluster of Orthologous Groups (A–Z)
KEGG_ko	KEGG Orthology number (K number)
KEGG_Pathway	KEGG pathway membership
GO_terms	Gene Ontology biological/molecular
PFAMs	Protein family domains
EC_number	Enzyme Commission number

```
emapper.py -i MAG1.faa -o MAG1_eggnog --cpu 8 --dmnd_db  
eggnog_proteins.dmnd
```

COG Functional Categories

J Translation, ribosomal structure	K Transcription
L Replication, recombination, repair	M Cell wall / membrane biogenesis
N Cell motility	O Post-translational modification
P Inorganic ion transport & metabolism	C Energy production & conversion
G Carbohydrate transport & metabolism	E Amino acid transport & metabolism
S Function unknown	

HUMAnN3: Community Functional Profiling

HUMAnN3 (HMP Unified Metabolic Analysis Network) profiles gene families and pathways from shotgun reads.

1

Taxonomic Profiling

MetaPhlAn4 identifies species present in the sample and their relative abundances. Used to select appropriate pangenome databases.

2

Translated Search (Tier 1)

Reads mapped to nucleotide pangenomes of identified species (ChocoPhlAn). Fast — only for known species.

3

Translated Search (Tier 2)

Unmapped reads searched against UniRef90 protein database using DIAMOND. Catches reads from uncharacterised organisms.

4

Gene Family Abundances

Outputs RPK (reads per kilobase) for each UniRef90 gene family. Stratified by species or unstratified (total community).

5

Pathway Reconstruction

Gene families mapped to MetaCyc and KEGG pathways using MinPath algorithm. Only complete/nearly complete pathways reported.

6

Normalisation

humann_renorm_table converts to relative abundance (relab) or copies per million (cpm). Enables cross-sample comparison.

```
humann --input reads.fastq.gz --output humann3_out/ --threads 8 --metaphlan-options "--bowtie2db metaphlan_db"
```

Antimicrobial Resistance (AMR) Gene Detection

Metagenomics enables surveillance of the resistome — all AMR genes in a microbial community.

CARD / RGI

card.mcmaster.ca

Comprehensive Antibiotic Resistance Database. RGI (Resistance Gene Identifier) searches protein sequences against CARD. Outputs: gene name, drug class, resistance mechanism, CVSS score.

```
rgi main -i MAG.faa -o rgi_output -t  
protein --clean
```

AMRFinderPlus

NCBI tool

NCBI tool for AMR gene detection in assemblies or proteins. Uses HMMs + BLAST against AMRFinder database. Identifies: AMR genes, virulence factors, stress response genes.

```
amrfinder -p MAG.faa -o amrfinder.tsv --  
plus --threads 8
```

ResFinder

*resfinder.cbs.dtu.
dk*

Webserver + standalone tool. Species-specific resistance predictions. Good for clinical isolates; also works on metagenome assemblies.

```
run_resfinder.py -i assembly.fasta -s  
"Other" -o resfinder/
```

Key output: gene name, drug class, % identity, mechanism of resistance (efflux / enzymatic / target modification).

SECTION 3

Mini-Project

Apply what you have learned to an independent dataset

Mini-Project: Introduction

You will work in groups of 2–3 to independently analyse a real soil shotgun metagenomic dataset you have NOT seen before. Apply the full pipeline from QC to functional analysis.

Dataset

Soil aggregates from a no-tillage upland field study (BioProject PRJDB18822). 3 no-till + 3 tilled samples (6 total). Goal: characterise microbial community differences driven by tillage.

Biological Question

How does tillage (tilled vs. no-till management) affect soil microbial community structure, diversity, and functional potential?

Groups

2–3 participants per group. All groups use the same 6-sample dataset (data/shotgun/soil/) — work through the full pipeline independently, then share and compare results at the end.

Time

~3–4 hours. Work systematically through the pipeline. You do NOT need to complete every analysis — prioritise quality over quantity.

Mini-Project: Tasks & Pipeline

1 QC & Preprocessing

Required

Run FASTQC → fastp → SEQKIT stats. Remove host reads (soil: no host removal needed). Summarise in MultiQC report.

2 Taxonomic Profiling

Required

Classify reads with Kraken2 + Bracken. Generate phylum/genus-level abundance table. Compare no-till vs. tilled.

3 Diversity Analysis

Required

Import to R (phyloseq or manual). Compute alpha diversity (Shannon, Chao1). Beta diversity PCoA (Bray-Curtis). PERMANOVA test.

4 Assembly & QC

Bonus

Assemble one sample with MEGAHIT (suggested: soil_DRR628691). Filter contigs ≥ 1000 bp. Run QUAST. Report N50 and contig counts.

5 Functional Annotation

Required

Run HUMAnN3 on one sample per condition (too slow for all). Compare pathway abundances. Which pathways differ between no-till and tilled soil?

6 MAG Recovery

Bonus

(Bonus) Bin assembled reads. Run CheckM2 + GTDB-Tk on ≥ 3 bins. Annotate with Prokka + eggNOG.

Mini-Project: Deliverables & Group Presentation

Deliverables

1

Brief written report (1–2 pages)

Methods used, key results (with figures), biological interpretation of findings. What is different between tillage treatments (no-till vs. tilled)?

2

All analysis scripts

Bash scripts and R Markdown saved to GitHub. Scripts must be reproducible — a classmate should be able to re-run them.

3

Figures (≥ 3 publication-quality)

Required: diversity plot (alpha or beta), taxonomy composition, one functional plot. Axis labels, legends, and informative titles.

4

5-minute group presentation

Slides summarising your findings. Cover: question → approach → results → interpretation. All group members present.

Evaluation Criteria

Scientific accuracy

30%

Correct interpretation of results. Appropriate statistical tests. No overinterpretation.

Reproducibility

25%

Scripts well-documented and reproducible. Clear README. Tools and parameters stated.

Data visualisation

25%

Informative, publication-quality figures. Colour, labels, scales appropriate.

Presentation clarity

20%

Clear message. Time management. Each member contributes.



Congratulations!

You have completed the 5-Day Metagenomics Course

Day 1	16S rRNA amplicon • QIIME2 • DADA2 • EMU • Databases
Day 2	Alpha & beta diversity • Ordination • PERMANOVA • ANCOM-BC
Day 3	Shotgun QC • Host removal • Kraken2 • MEGAHIT • QUAST
Day 4	Binning • MetaBAT2 • CheckM2 • GTDB-Tk taxonomy
Day 5	Prokka • eggNOG • HUMAnN3 • AMR genes • Mini-project

Next steps: publish your results • explore Galaxy/QIIME2 workflows • contribute to GitHub repositories

KEY REFERENCES

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Prokka: rapid prokaryotic genome annotation
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CARD 2023: expanded curation and resistome prediction
Nucleic Acids Research 51: D690–D699

Galperin et al. (2021)

COG database update: focus on microbial diversity and widespread pathogens
Nucleic Acids Research 49: D274–D281

Gurevich et al. (2013)

QUAST: quality assessment tool for genome assemblies
Bioinformatics 29: 1072–1075